



3.5 Vector applications to kinematics (HL)

Geometry and trigonometry · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A moving particle's position vector $\mathbf{r}(t)$ gives its location at time t . Differentiating componentwise gives the velocity vector $\mathbf{v}(t)=d\mathbf{r}/dt$, and differentiating again gives acceleration $\mathbf{a}(t)=d\mathbf{v}/dt$; speed is $|\mathbf{v}(t)|$. For constant velocity, $\mathbf{r}(t)=\mathbf{r}_0+t\cdot\mathbf{v}$. Two moving objects' relative position vector ($\mathbf{r}_B-\mathbf{r}_A$) can be used to find the distance between them at a given time, or to test whether they ever collide (occupy the same position at the same instant).

Worked example: follow the reasoning

A particle's position vector is $\mathbf{r}(t)=(3+2t)\mathbf{i}+(5-t^2)\mathbf{j}$ (metres), t in seconds. Find (a) the velocity vector $\mathbf{v}(t)$, (b) the velocity and speed at $t=2$ s.

Solution: (a) $\mathbf{v}(t)=2\mathbf{i}-2t\mathbf{j}$. (b) $\mathbf{v}(2)=2\mathbf{i}-4\mathbf{j}$; speed= $|\mathbf{v}(2)|=\sqrt{4+16}=\sqrt{20}\approx 4.47$ m/s.

Quick check · multiple choice

A particle has constant velocity vector $\mathbf{v}=(3,-4)$ m/s. What is its speed?

- A 1 m/s
- B 5 m/s
- C 7 m/s
- D 25 m/s

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Two boats' positions (km, relative to a harbour) at time t hours are given by [8]
 $\mathbf{r}_A(t)=(2+3t, 1+4t)$ and $\mathbf{r}_B(t)=(10-t, 2+2t)$. (a) Find the velocity vector of each boat. (b) Find the vector $\mathbf{AB}$ (from boat A to boat B) in terms of t . (c) Find the distance between the boats at $t=1$ hour. (d) Determine, with justification, whether the boats will ever collide.

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $\mathbf{v}_A = (3, 4)$, $\mathbf{v}_B = (-1, 2)$. (b) $\mathbf{AB} = \mathbf{r}_B - \mathbf{r}_A = (8 - 4t, 1 - 2t)$. (c) At $t = 1$: $\mathbf{AB} = (4, -1)$, distance $= \sqrt{16 + 1} = \sqrt{17} \approx 4.12$ km. (d) Collision requires $8 - 4t = 0$ ($t = 2$) and $1 - 2t = 0$ ($t = 0.5$) simultaneously; since these give different values of t , the boats never occupy the same position at the same time, so they do not collide.

Quick check answer

B. Speed $= |\mathbf{v}| = \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5$ m/s.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.